Vietnam Portfolio Risk and Stress Report

FUESSV30 reference portfolio | VND100 billion shadow mandate | Information cutoff 30 September 2026

Decision

For a benchmark-tracking mandate, retain the unhedged exposure and approve a time-limited Expected Shortfall exception. For an absolute-risk mandate with a binding 12% ES limit, reduce beta toward 0.70 only after futures execution controls are completed.

1-day 99% VaR	10-day 97.5% ES	Backtest	Beta to VN30
3.39%	**12.28%**	**2 / 250**	**0.99**

The standard historical VaR limit passes by only 11 basis points. The 10-day ES limit fails by 28 basis points. More importantly, the normal model understates downside risk: normal VaR is 2.87%, while a skew and kurtosis adjusted Cornish-Fisher VaR is 4.16%.

Investment outcome

Mandate	Decision	Why
Benchmark tracking	Retain unhedged	A beta hedge raises tracking error from 2.35% to 6.51%.
Absolute risk	Hedge toward beta 0.70	The overlay reduces 99% historical VaR from 3.39% to 2.46%.

This distinction is central. Hedging is not automatically superior; the correct action follows the mandate objective and approved risk budget.

Portfolio and data lineage

Portfolio construction

The model freezes the 30 September 2026 SSIAM VN30 ETF holdings and rescales them to a VND100 billion analytical mandate. Reported weights sum to 100.02% because of disclosure rounding; the analytical layer normalises them to 100.00% while retaining reported weights for audit.

Layer	Source	Use	Control
Holdings	SSIAM September 2026 factsheet	Names, weights, sectors	Frozen as-of date and page reference
Equities	VNDirect Finfo	Adjusted prices and traded values	Coverage and missing-quote audit
Benchmark	VNDirect Finfo	VN30 return series	Common cutoff and calendar
Futures	HNX	Multiplier and contract terms	Index-equivalent only until contract selected
Backtest	BIS MAR99	Traffic-light reference	Governance reference, not capital compliance

Short-history policy

TCX uses a Financials sector proxy excluding TCX in both the 500-day and 250-day models. VPL uses a Consumer Discretionary proxy excluding VPL in the 500-day model and its own history in the 250-day model. Historical replays before listing use explicit sector proxies. No substitution is silent.

Interpretation boundary

All historical results are current-weight replays. They answer how today's portfolio would have behaved under prior return environments. They are not reconstructed FUESSV30 fund returns and contain survivorship and composition bias.

Risk profile and model challenge

Measure	500-day	250-day	Interpretation
Annualised volatility	20.19%	20.59%	Stable long and recent estimate
EWMA volatility	14.21%	14.17%	Current regime is calmer
Historical VaR 99%	3.39%	3.25%	Empirical tail
Normal VaR 99%	2.87%	2.98%	Understates non-normal tail
Cornish-Fisher VaR 99%	4.16%	3.32%	Higher-moment challenge
Skewness	-0.80	-0.68	Negative left-tail asymmetry
Excess kurtosis	7.87	4.00	Normal distribution equals zero

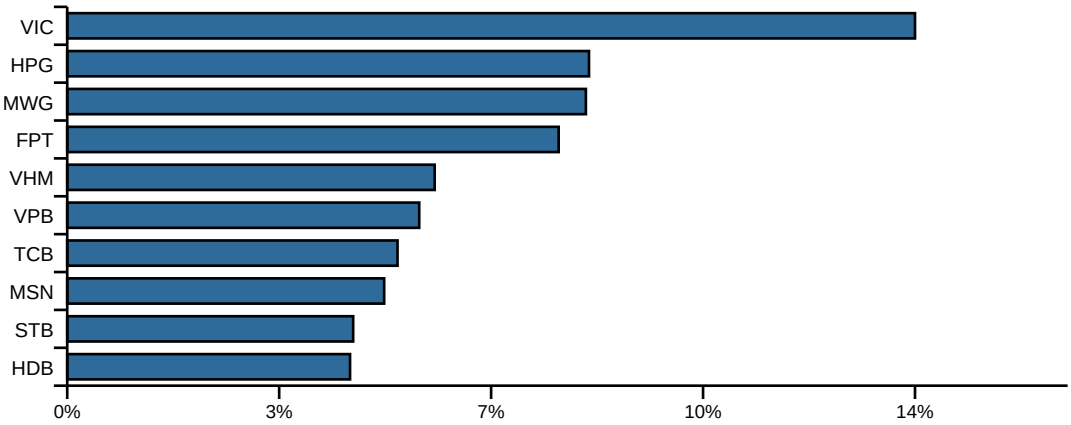
The distribution challenge changes the decision. A normal covariance model appears comfortably inside the 3.50% limit. Historical VaR is near the boundary, while Cornish-Fisher VaR breaches it. Historical ES and scenario loss should therefore drive the risk action.

Regime and diversification

Average pairwise correlation is 0.37 over 500 days and 0.35 over the latest 60 days. The diversification ratio is 1.64. The current regime is calmer and is not exhibiting a broad correlation spike, but the historical left tail remains material.

Risk attribution and concentration

Top contributors to portfolio volatility



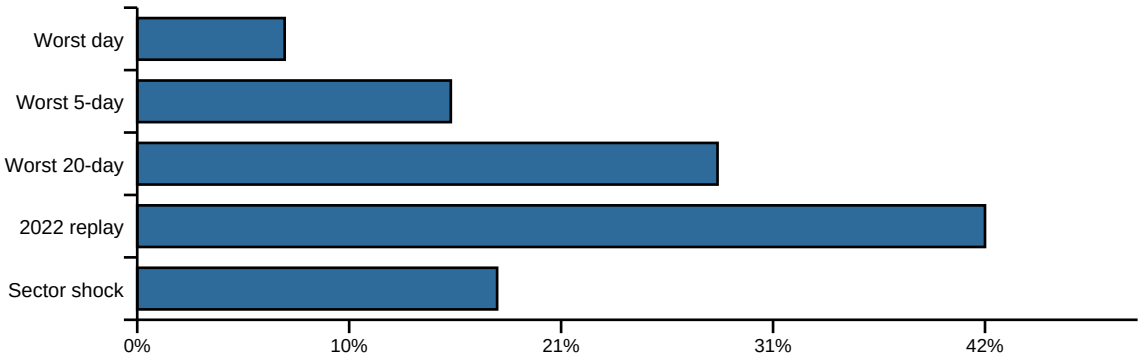
Name	Weight	Share of risk	Risk / weight
VIC	10.62%	13.57%	1.28x
HPG	8.72%	8.35%	0.96x
MWG	6.84%	8.30%	1.21x
FPT	8.62%	7.87%	0.91x
VHM	4.37%	5.88%	1.35x
VPB	4.72%	5.64%	1.19x
TCB	4.65%	5.29%	1.14x

VIC contributes 13.57% of portfolio volatility from a 10.62% weight. Real Estate contributes 21.90% of risk from a 16.88% weight, the clearest sector-level risk concentration.

Stress testing

Scenario	Loss	VND impact	Use
Worst historical day	-7.23%	VND -7.2bn	2020-03-09
Worst historical 5-day	-15.37%	VND -15.4bn	2025-04-02 to 2025-04-09
Worst historical 20-day	-28.44%	VND -28.4bn	2020-03-04 to 2020-03-31
2022 deleveraging replay	-41.55%	VND -41.6bn	Current-weight replay
Hypothetical sector shock	-17.64%	VND -17.6bn	Defined sector vector
Correlation-one 99% VaR	-4.85%	VND -4.9bn	Diversification failure

Scenario loss as a share of NAV



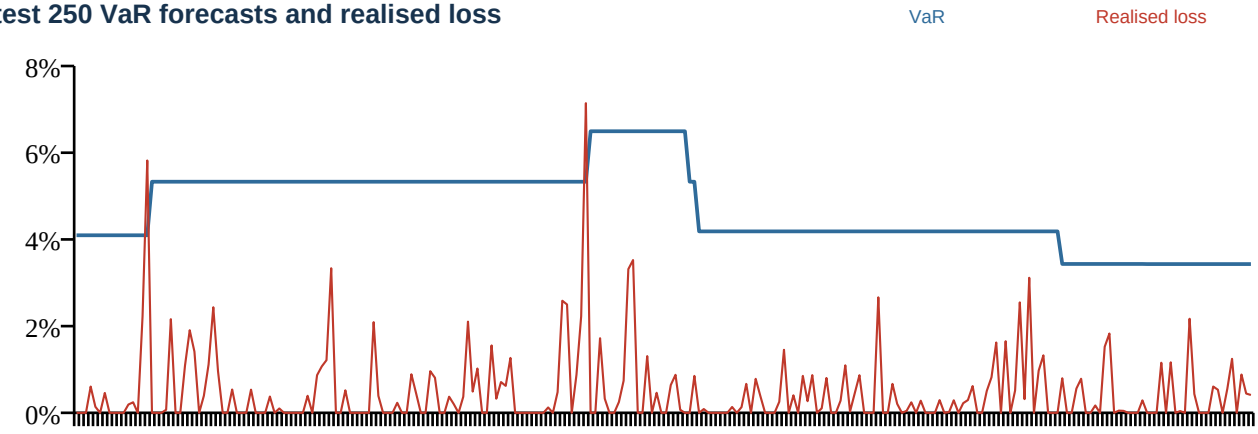
Reverse stress

A simplified beta-only reverse stress reaches a 15% NAV loss at an estimated VN30 shock of -15.18%. This is a threshold diagnostic, not a full nonlinear scenario.

Backtesting and validation

Control	Result	Interpretation
Forecast observations	250	Latest rolling forecasts
Exceptions	2	2 realised losses exceeded prior-day VaR
Traffic-light zone	GREEN	Green: 0 to 4 exceptions
Observed exception rate	0.80%	Expected rate is 1.00%
Kupiec POF p-value	0.742	Do not reject unconditional coverage at 5%

Latest 250 VaR forecasts and realised loss



Validation conclusion

The model passes the traffic-light and unconditional-coverage tests. This supports the frequency of breaches, not the independence of exceptions or correctness of sector proxies. A production framework would add independence and conditional-coverage tests, benchmark reconciliation and formal model-change governance.

Hedge analysis

Measure	Unhedged	Beta 0.70 hedge	Change
Annualised volatility	20.19%	14.40%	-5.79%
Historical VaR 99%	3.39%	2.46%	-0.93%
Historical ES 97.5%	4.15%	2.97%	-1.18%
Tracking error	2.35%	6.51%	4.16%
Beta	0.99	0.70	-0.29

The index-equivalent position is 151 short contracts at a VND 190.7 million notional per contract. Estimated VaR falls 27.41% and volatility falls 28.66%. Tracking error rises by 4.16%.

Execution gates

Gate	Required before trade
Contract	Select maturity and use actual settlement price
Basis	Measure and limit cash-futures basis risk
Liquidity	Insert current initial margin and stressed variation margin
Roll	Define roll window, transaction cost and slippage
Governance	Approve tracking-error impact and rebalance tolerance

Governance, limitations and sources

Material limitations

- Frozen current weights introduce survivorship and composition bias in historical replays.
- Sector proxies suppress issuer-specific risk for short-history securities.
- Carry-forward creates zero returns on missing quote days and concentrates the move on the next quote.
- Square-root-of-time ES scaling is a limit approximation, not a full 10-day simulation.
- Cornish-Fisher VaR can be unstable when sample higher moments are extreme.
- Liquidity excludes stressed volume decay, queues at price limits and nonlinear market impact.
- The hedge omits actual basis, maturity, margin, roll, tax and execution costs.

Primary sources

SSIAM September 2026 ETF factsheet: https://ssiam.com.vn/upload/files/QuanLyQuy/ETF%20Monthly%20Report%20September%202026.pdf

SSIAM VN30 fund page: https://ssiam.com.vn/en/fund-information-VN30

HNX VN30 futures specification: https://www.vpdt.hnx.vn/vi-vn/huong-dan/chi-tiet-thu-tuc-36-65.html

BIS Basel Framework MAR99: https://www.bis.org/committees/bcbs/basel-framework/standard/mar/99/inforce/2019-12-15/published/2019-12-15

VNDirect Finfo public API: https://api-finfo.vndirect.com.vn/v4

Reproducibility

The repository contains the frozen data snapshot, dependency-light Python risk engine, CSV outputs, formula-driven Excel workbook, model card and independent checks. Re-fetching market data creates a new evidence set and should be versioned as a new model run.